

Active Perceptual Inference: A Corticothalamic-Inspired Dynamic Nested Recurrent Network for Multimodal Sentiment Analysis with Incomplete Data

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Abstract

Random frame-level data missing is a critical challenge in multimodal sentiment analysis. Existing methods are largely limited to passive completion via single-pass feed-forward connections and static cross-modal fusion, which struggle to generate high-quality completed features. However, the brain is not a passive recipient of external information but a dynamic system for active perceptual inference. Its core lies in the dynamic nested recurrences formed by intra-cortical recurrent completion mechanisms and corticothalamic circuits, which iteratively perform perceptual inference. Inspired by this, we propose the Dynamic Nested Recurrent Network (DNRNet). It is the first to introduce recurrent inference into the data completion task, achieving a paradigm shift from passive completion to active perceptual inference. Its local recurrent loop simulates intra-cortical recurrent pattern completion to perform perceptual inference and generate local correction features. The global recurrent loop simulates the modulatory function of the thalamus, calculating modality confidence to dynamically weight and integrate cross-modal information, generating global correction features. The local and global correction features are fused to obtain the completion signal, which is then combined with the input features of the current iteration to serve as the input for the next iteration. Experiments on the MOSI, MOSEI, and SIMS datasets demonstrate that DNRNet achieves an average accuracy improvement of 1.5%–2.0% over baseline models across all missing rates, validating the superiority of the brain-inspired approach in complex missing data scenarios.

1. Introduction

Multimodal Sentiment Analysis (MSA) aims to achieve a more comprehensive understanding of human emotion

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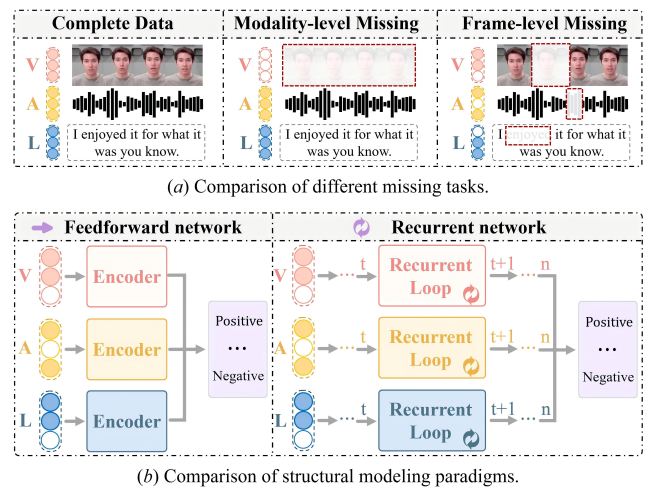


Figure 1. Comparison of missing tasks and modeling paradigms. This work addresses the **frame-level missing** task shown in (a) and adopts a novel **recurrent network** illustrated on the right in (b). This design represents a paradigm shift from passive feature completion to active perceptual inference.

by jointly modeling multi-source information such as language, audio, and visual data [5, 36]. However, in practical applications, the integrity of multimodal data faces challenges, often accompanied by varying degrees of missingness. Based on the granularity of the missing elements, existing research typically categorizes multimodal missing tasks into modality-level and frame-level missing scenarios, as illustrated in Figure 1(a). The former assumes that an entire modality is unavailable, such as the complete non-availability of vision or audio modalities [12, 23, 46]. In contrast, frame-level missingness focuses on a finer scenario where partial frames or segments are missing within a modality due to factors like noise, occlusion, or transmission instability [13, 45, 48]. This fine-grained missingness often occurs randomly and can co-exist across multiple modalities, leading to the fragmentation of emotional cues

and cross-modal inconsistency. Consequently, frame-level missing tasks are considered more complex. In this work, we focus on the frame-level missing task, aiming to develop effective sentiment modeling under incomplete data.

Recent studies have made significant progress in addressing frame-level missingness in MSA, with core strategies primarily relying on cross-modal consistency [13, 14, 45, 48]. For example, P-RMF [48] maps unimodal data into a Gaussian latent space to capture core features and enhances representations using a unidirectional feedforward dynamic injection mechanism. However, these methods tend to overlook modality-specific information, which limits their ability to effectively recover fine-grained intra-modality local semantics. Furthermore, they universally adopt a single-pass feedforward network for passive feature completion, as illustrated on the left side of Figure 1(b). This passive and unidirectional mechanism lacks self-adaptive correction capabilities, making it difficult to eliminate the redundancy and noise introduced during the completion process [19].

Neuroscience research indicates that the human brain does not passively accept sensory input when processing incomplete or ambiguous information. Instead, it performs active perceptual inference based on prior expectations to identify objects [21]. This process is jointly supported by intra-cortical recurrent pattern completion mechanisms and corticothalamic circuits [1, 22, 30], as conceptually illustrated in Figure 2(a). Within the cerebral cortex, initially incomplete information is forwarded bottom-up to higher-order cortical areas for inference, generating predictive signals. These predictions are then fed back top-down to lower-order cortical areas, activating pattern completion processes [21]. Simultaneously, the thalamus, as a key relay hub, participates in the perceptual completion process by regulating and integrating multi-sensory information via the corticothalamic circuit [30]. The completed features are forwarded again to higher-order cortices for the next round of perceptual inference. This constantly circulating information flow enables the brain to quickly and accurately identify objects even under incomplete sensory conditions [21].

Inspired by this brain function, we propose the Dynamic Nested Recurrent Network (DNRNet), which is composed of local and global recurrent loops. To the best of our knowledge, it is the first model to introduce recurrent perceptual inference into the task of missing data completion, representing a paradigm shift from passive completion to active perceptual inference, as illustrated on the right side of Figure 1(b). The **Local Recurrent Loop** simulates the recurrent pattern completion mechanism between higher- and lower-order cortical areas. Specifically, the High-Level Inference Module simulates the higher-order cortex by inferring local segments within a single modality and generating feedback signals. The Local Perception Module simulates

the lower-order cortex. It is responsible for receiving these feedback signals and producing local correction features, which subsequently drive the extraction of global correction information in the global recurrent loop. This local recurrent mechanism enables DNRNet to effectively restore fine-grained intra-modality semantics based on the modality's own prior expectations. The **Global Recurrent Loop** simulates the corticothalamic circuits. The Confidence-Aware Module within it simulates the thalamus's regulatory function. It estimates modality reliability to guide the Global Perception Module in dynamically integrating information from other modalities and generating global correction features. The local and global correction features are then combined, and this combined signal is forwarded to the high-level inference module, where it is merged with the current iteration's input features to serve as the input for the next recurrence. This nested recurrent design enables adaptive refinement of input features, reducing redundancy and noise.

Our contributions can be summarized as follows:

- Inspired by the brain's active perceptual inference, we are the first to introduce a recurrent inference framework into the task of multimodal missing data completion, achieving a paradigm shift from passive completion to active perceptual inference.
- We propose a dynamic nested recurrent network. The local loop simulates intra-cortical recurrent pattern completion mechanisms to recover fine-grained intra-modal semantics, while the global loop emulates the corticothalamic circuit. This nested recurrent architecture enables the model to adaptively correct input information.
- Extensive experiments demonstrate that DNRNet outperforms state-of-the-art baselines, achieving an average accuracy improvement of 1.5%–2.0% across all missing rates, validating the effectiveness of our approach.

2. Related Work

2.1. MSA with Incomplete Data

Depending on the scope of the missing information, missing data in Multimodal Sentiment Analysis (MSA) can generally be categorized into two types. The first is modality-level missingness, where an entire modality is unavailable. To address this issue, existing studies have proposed generative approaches [4, 6, 15, 18, 33, 47] and joint learning methods [9, 16, 17, 20, 42, 43]. The second type is frame-level missingness. Since the missing information occurs in fine-grained local fragments, this problem is more challenging. To tackle this, one line of work employs feature reconstruction strategies [24, 34, 39]. For example, EMT-DLFR [24] addresses fine-grained intra-modal data loss by reconstructing missing features and minimizing the representation gap between complete and incomplete views. Another paradigm employs low-rank tensor completion tech-

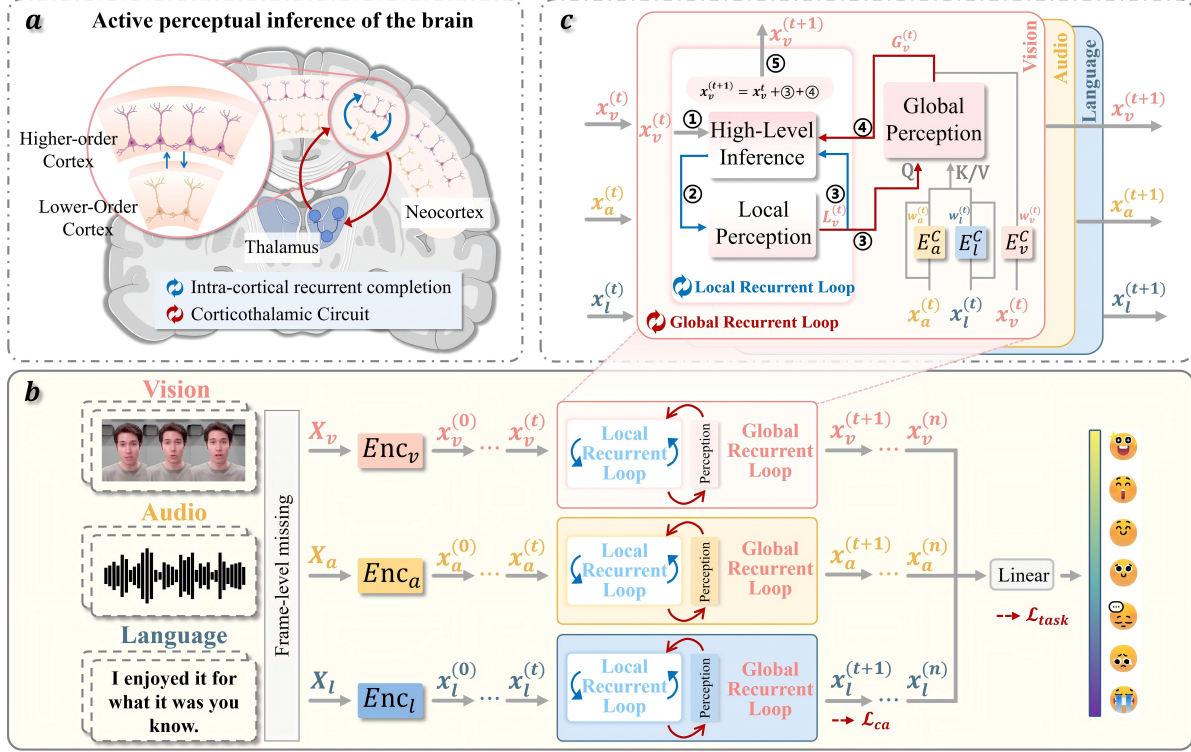


Figure 2. **An overview of the DNRNet model architecture.** (a) Active perceptual inference of the brain: the intra-cortical recurrent pattern completion mechanism and the corticothalamic circuit. (b) The overall structure of DNRNet, which undergoes n iterations of nested recurrent processing. (c) Detailed processing flow of the t -th iteration, using the vision modality as an example to highlight the specific interaction between the Local and Global Recurrent Loops.

niques to reconstruct missing fragments from a mathematical optimization perspective [10, 26, 29]. A third direction focuses on learning unified cross-modal representations [13, 40, 48]. For instance, P-RMF [48] learns a robust cross-modal proxy representation via Gaussian latent mapping and uncertainty quantification. Crucially, unlike existing passive feedforward models, our DNRNet introduces a brain-inspired dynamic nested recurrent architecture to perform active perceptual inference, enabling self-adaptive correction for the robust recovery of incomplete multimodal semantics.

2.2. Brain-Inspired Approaches

The brain is the most complex and sophisticated dynamic information processor to have evolved in nature [11]. In recent years, there has been growing interest in drawing inspiration from neuroscience to develop more efficient and biologically interpretable algorithmic models. Inspired by cognitive neuroscience, recent studies have begun to explore brain-inspired architectures for multimodal perception [25, 31, 32, 35], aiming to emulate key mechanisms underlying multisensory integration, attention regulation, and robust perception. For instance, BIG-FUSION [35] models the brain’s parallel processing mechanisms and proposes

an aligned global–local fusion framework that allows two types of information to be simultaneously captured and preserved. However, existing brain-inspired methods primarily focus on cross-modal integration or attention control. The brain is an efficient and active perceptual inference system, capable of maintaining stable perception and decision-making even under incomplete information. In this work, we further draw inspiration from the brain’s corticothalamic circuits to propose an architecture capable of performing iterative inference for addressing missing data in multimodal sentiment analysis.

3. The Proposed Method

3.1. Problem Definition

We begin by formally defining the complete multimodal data sequence. Given a video clip with modalities $m \in \{v, a, l\}$, the intact sequence is denoted as $U_m \in \mathbb{R}^{T_m \times D_m}$, where T_m is the sequence length and D_m is the feature dimensionality. However, our focus is on the challenge of frame-level missingness, a form of intra-modal partial data loss. Such missingness, often caused by physical constraints like occlusion or channel interruptions, demands fine-grained handling. In this study, the missing data $X_m \in$

$\mathbb{R}^{T_m \times D_m}$ is modeled by introducing randomly missing features at the frame level within modality sequences.

3.2. Overall Framework

The proposed DNRNet is illustrated in Figure 2. Inspired by the intra-cortical recurrent pattern completion mechanism and the corticothalamic circuit (as shown in Figure 2(a)), we design a nested recurrent network for each modality that runs for n iterations. The overall framework is presented in Figure 2(b). The proposed DNRNet consists of two loops: the Local Recurrent Loop and the Global Recurrent Loop. To detail our recurrent methodology, we analyze the t -th iteration of the vision modality as a specific example, which is detailed in Figure 2(c). The process is as follows: First (**Local Recurrent Loop**): The High-Level Inference Module (HLI) in the local recurrent loop receives the feature sequence $x_m^{(t)}$ from the Unimodal Encoder and generates inference signals. Next, the Local Perception Module (LP) takes these inference signals as input and produces the local correction features $L_m^{(t)}$. Then (**Global Recurrent Loop**): The Confidence-Aware Module (CA) estimates the reliability of each modality. Guided by these confidence scores, the Global Perception Module (GP) dynamically weights and aggregates features from other modalities to produce the global correction features $G_m^{(t)}$. Finally (**Iterative Update**): $L_m^{(t)}$, $G_m^{(t)}$, and $x_m^{(t)}$ are fused within the high-level inference module, and the result serves as the input for the next iteration, denoted as $x_m^{(t+1)}$. During training, a missing-reconstruction loss is employed as an auxiliary supervisory signal to further enhance the model’s ability to recover incomplete data.

3.3. Unimodal Encoder

We construct modal encoders $Enc_m(\cdot)$ for vision, audio, and language separately to extract and unify the modality representations. Specifically, the vision and audio modality encoders are composed of linearly transformed and Transformer encoder layers. For the language modality, semantic representations are first extracted using the pre-trained language model BERT, which are then fed into a Transformer layer to capture richer contextual dependencies. The final dimensionally consistent feature $x_m^{(0)}$, which serves as the input to the subsequent modules, is obtained as follows:

$$x_m^{(0)} = Enc_m(X_m). \quad (1)$$

3.4. Local Recurrent Loop

Prior research neglects modality-specific information, which severely limits its ability to recover intra-modal local semantics effectively. In contrast, the recurrent pattern completion mechanism within the cerebral cortex drives the brain to perform active perceptual inference [21]. This inherent self-correction mechanism enables the brain to dy-

namically refine local semantic information. Inspired by this self-correction process, we design the Local Recurrent Loop, which primarily consists of two processes: bottom-up feedforward and top-down feedback.

First, the current feature $x_m^{(t)}$ is passed through a feedforward projection (FF_m) and fed into the High-Level Inference Module (E_m^{HLI}). This module, composed of a two-layer Transformer encoder, simulates the higher-order cortex, performing deep inference over the intra-modal features and generating a local inference signal $g_m^{(t)} \in \mathbb{R}^{T_m \times D_m}$:

$$g_m^{(t)} = E_m^{HLI}(FF_m(x_m^{(t)})). \quad (2)$$

Subsequently, $g_m^{(t)}$ is mapped via a feedback projection (Feedback _{m}) to the Local Perception Module (E_m^{LP}). This module, composed of a two-layer MLP with a Tanh activation function, simulates the lower-order cortex, producing the local correction features $L_m^{(t)} \in \mathbb{R}^{T_m \times D_m}$:

$$L_m^{(t)} = E_m^{LP}(\text{Feedback}_m(g_m^{(t)})), \quad (3)$$

where $L_m^{(t)}$ represents the expected pattern of modality m . It serves as the Query for the attention mechanism driving the global recurrent loop.

3.5. Global Recurrent Loop

While the Local Recurrent Loop addresses the recovery of modality-specific information and achieves self-correction, relying solely on intra-modal self-inference can easily lead to ambiguities in local semantic prediction. Such a lack of calibration from global cross-modal cues negatively impacts the robustness of the final features. To overcome this limitation, we design the Global Recurrent Loop. It simulates the function of the corticothalamic circuit to realize the dynamic modulation of cross-modal information and achieve global consistency correction.

First, the Confidence-Aware Module (E_m^C) emulates the thalamic modulatory function over multisensory information flow. It is designed to estimate the reliability of the current feature $x_m^{(t)}$. We employ a Transformer structure to perceive the sequential features and a linear classifier to generate the confidence score $w_m^{(t)}$ for modality m :

$$w_m^{(t)} = E_m^C(x_m^{(t)}), \quad (4)$$

where $w_m^{(t)} \in [0, 1]$ is obtained via a Sigmoid activation.

To guide the model in learning meaningful confidence scores, we introduce the Confidence-Aware Loss ($\mathcal{L}_{ca}$) as an auxiliary supervision. We compute the Mean Squared Error (MSE) loss between the predicted score w_m^k and the soft ground-truth label $\hat{w}_m^k$. The overall confidence-aware loss is defined as:

$$\mathcal{L}_{ca} = \frac{1}{N_b \cdot M} \sum_{k=1}^{N_b} \sum_{m \in \{l, v, a\}} (w_m^k - \hat{w}_m^k)^2, \quad (5)$$

where N_b denotes the size of the current training batch, $M = 3$ is the number of modalities, and w_m^k denotes the predicted confidence score for modality m in the k -th sample. The soft ground-truth is defined as $\hat{w}_m^k = 1 - r_m^k$, where r_m^k represents the missing ratio.

Then, the global perception module achieves global consistency correction through cross-modal attention. Specifically, we aggregate features from other modalities i , weighted by their confidence scores $w_i^{(t)}$, to generate the cross-modal context $C_m^{(t)}$:

$$C_m^{(t)} = \sum_{i \neq m} w_i^{(t)} x_i^{(t)}, \quad (6)$$

where $C_m^{(t)}$ serves as the Key and Value for the attention mechanism.

Subsequently, the $L_m^{(t)}$ output by the local recurrent loop is used as the Query to retrieve features from the cross-modal context information, and obtain the global correction feature $G_m^{(t)} \in \mathbb{R}^{T_m \times D_m}$ by multiplying it with the residual weight $(1 - w_m^{(t)})$ of the current modality:

$$G_m^{(t)} = (1 - w_m^{(t)}) \cdot \text{Attention}(L_m^{(t)}, C_m^{(t)}, C_m^{(t)}). \quad (7)$$

3.6. Iterative Update and Convergence Mechanism

Iterative Update. We sum the local correction feature $L_m^{(t)}$ and the global correction feature $G_m^{(t)}$ to obtain the unified correction signal. This signal is then integrated into the current feature $x_m^{(t)}$ using a trainable scaling factor λ :

$$x_m^{(t+1)} = \text{LayerNorm}_m \left(x_m^{(t)} + \lambda \cdot (L_m^{(t)} + G_m^{(t)}) \right). \quad (8)$$

This nested recurrent architecture ensures continuous iterative refinement and self-adaptive correction of the features, effectively suppressing redundancy and noise.

Convergence Mechanism. To ensure both the computational efficiency and iterative stability of the model, we design a convergence-driven recurrent mechanism. After the end of each iteration t , we check whether the average relative L_2 norm change $\Delta_{rel}^{(t+1)}$ across all modality features falls below a predefined threshold ϵ :

$$\Delta_{rel}^{(t+1)} = \frac{1}{N_b} \sum_{k=1}^{N_b} \frac{\|S_k^{(t+1)} - S_k^{(t)}\|_2}{\|S_k^{(t)}\|_2 + \xi}, \quad (9)$$

where $S^{(t)}$ denotes the aggregate state vector obtained by concatenating the flattened feature sequences across all modalities, and ξ is a small constant introduced to prevent division by zero. If $\Delta_{rel}^{(t+1)} < \epsilon$, the loop is terminated, thereby ensuring the model runs for only the minimum number of iterations required for the feature space to stabilize. In this paper, the ϵ is set to 0.0001. This guarantees the quality of the feature completion and enhances the model's iterative stability.

3.7. Fusion and Reconstruction

Fusion and Sentiment Prediction. Upon completion of the recurrent process, we obtain the final completed features $x_m^{(\text{final})}$. For the final sentiment prediction, we first fuse the completed features of all modalities:

$$H = x_v^{(\text{final})} + x_a^{(\text{final})} + x_l^{(\text{final})}. \quad (10)$$

Subsequently, the fused feature H is compressed into a fixed-length aggregated vector using global average pooling. This vector is finally fed into a linear classifier for sentiment prediction $\hat{y} = \text{Linear}(H)$. To optimize the model, we apply the MSE loss between the predicted score $\hat{y}$ and the ground-truth sentiment label y :

$$\mathcal{L}_{\text{task}} = \frac{1}{N_b} \sum_{k=1}^{N_b} (y^k - \hat{y}^k)^2. \quad (11)$$

Feature Reconstruction. To enhance the robustness and fidelity of the cyclically completed features, we introduce the feature reconstruction loss as an auxiliary supervisory signal. We devise an independent Reconstructor (E_m^{Rec}) for each modality, consisting of two Transformer encoder layers. The reconstruction process is expressed as $\hat{x}_m = E_m^{\text{Rec}}(x_m)$. The reconstruction loss $\mathcal{L}_{\text{rec}}$ is defined as the squared L_2 loss between the reconstructed feature $\hat{x}_m$ and the complete input feature $u_m = \text{Enc}_m(U_m)$:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N_b} \sum_{k=1}^{N_b} \|u_m^k - \hat{x}_m^k\|_2^2, \quad (12)$$

where N_b denotes the batch size.

3.8. Objective optimization

In summary, the model is optimized via a weighted combination of three losses: the sentiment prediction loss $\mathcal{L}_{\text{task}}$, confidence-aware loss $\mathcal{L}_{ca}$, and reconstruction loss $\mathcal{L}_{\text{rec}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \alpha \cdot \mathcal{L}_{ca} + \beta \cdot \mathcal{L}_{\text{rec}}, \quad (13)$$

where α and β are hyperparameters that balance the contributions of the corresponding losses.

4. Experiments

4.1. Datasets

We performed a comprehensive experimental analysis on three benchmark datasets: **MOSI** [41] consists of 2,199 opinion video clips from 93 YouTube movie reviews, each annotated with a sentiment score ranging from -3 (strongly negative) to 3 (strongly positive), with aligned visual, audio, and language modalities. It is split into 1,284 training, 229 validation, and 686 testing samples. **MOSEI** [2] contains 22,856 video clips with annotations from -3 (strongly

Table 1. Comparison of model performance on MOSI and MOSEI. Note: The lower MAE corresponds to superior results.

Method	MOSI						MOSEI					
	Acc-7	Acc-5	Acc-2	F1	MAE	Corr	Acc-7	Acc-5	Acc-2	F1	MAE	Corr
MISA	29.85	33.08	71.49 / 70.33	71.28 / 70.00	1.085	0.524	40.84	39.39	71.27 / 75.82	63.85 / 68.73	0.780	0.503
Self-MM	29.55	34.67	70.51 / 69.26	66.60 / 67.54	1.070	0.512	44.70	45.38	73.89 / 77.42	68.92 / 72.31	0.695	0.498
MMIM	31.30	33.77	69.14 / 67.06	66.65 / 64.04	1.077	0.507	40.75	41.74	73.32 / 75.89	68.72 / 70.32	0.739	0.489
CENET	30.38	37.25	71.46 / 67.73	68.41 / 64.85	1.080	0.504	47.18	47.83	74.67 / 77.34	70.68 / 74.08	0.685	0.535
TETFN	30.30	34.34	69.76 / 67.68	65.69 / 63.29	1.087	0.507	30.30	47.70	69.76 / 67.68	65.69 / 63.29	1.087	0.508
TFR-Net	29.54	34.67	68.15 / 66.35	61.73 / 60.06	1.200	0.459	46.83	34.67	73.62 / 77.23	68.80 / 71.99	0.697	0.489
ALMT	30.30	33.42	70.40 / 68.39	72.57 / 71.80	1.083	0.498	40.92	41.64	76.64 / 77.54	77.14 / 78.03	0.674	0.481
LNLN	34.26	38.27	72.55 / 70.94	72.73 / 71.25	1.046	0.527	45.42	46.17	76.30 / 78.19	77.77 / 79.95	0.692	0.530
P-RMF	34.19	38.50	72.81 / 71.53	72.93 / 71.69	1.038	0.525	44.63	45.87	78.14 / 78.83	79.33 / 80.39	0.658	0.589
DNRNet	35.36	39.11	74.22 / 73.19	74.24 / 73.28	1.036	0.538	46.89	47.84	78.28 / 78.89	78.74 / 80.18	0.655	0.590

negative) to 3 (strongly positive). It is divided into 16,326 training, 1,871 validation, and 4,659 testing samples, and includes aligned visual, audio, and language modalities. **SIMS** [37] is a Chinese multimodal sentiment benchmark comprising 2,281 video clips, annotated from -1 (negative) to 1 (positive). It includes 1,368 training, 456 validation, and 457 testing samples.

4.2. Missingness Settings and Evaluation Metrics

Missingness Settings. For the visual and audio modalities, missing segments are filled with zero vectors, while in the language modality, missing segments are replaced with [UNK] tokens, as used in BERT [3]. During training, a random missing rate is independently sampled for each sample in each modality, followed by a Bernoulli process applied to valid positions to determine which frames are masked. To prevent overfitting to missing data, half of the samples for each modality are assigned a zero missing rate. During testing, missing rates range from 0 to 0.9, with a step size of 0.1, and missing positions are randomly sampled independently for each modality.

Evaluation Metrics. For MOSI and MOSEI, we report the seven-class accuracy (Acc-7), five-class accuracy (Acc-5), binary classification accuracy (Acc-2), and the F1 score. We utilize two common schemes for Acc-2 and F1 calculation: negative/positive (left-side value of /) and negative/non-negative (right-side value of /). Additionally, performance is quantified using the Mean Absolute Error (MAE) and the Pearson correlation coefficient (Corr). The evaluation on the SIMS dataset includes Acc-5, Acc-2, three-class accuracy (Acc-3), F1 score, MAE, and Corr. All reported metrics, with the exception of the MAE, prioritize higher values as indicators of superior model performance.

4.3. Implementation Details

Our model is implemented using the PyTorch framework and trained on an NVIDIA L40S GPU, employing the

AdamW optimizer for parameter updates. To ensure robustness and reliability, all experiments are repeated with three different random seeds, and the average results are reported. Training utilized a fixed schedule of 200 epochs across all datasets, incorporating warm up, cosine annealing, and early stopping strategies. The input vector length is set to 8, the hidden dimension to 128, and the batch size to 64, with an initial learning rate of 1×10^{-4} . The High-Level Inference Module E_m^{HLI} utilizes two Transformer encoder layers with 8 heads and a hidden dimension of 128. The Confidence-Aware Module E_m^C is configured with 8 heads, hidden dimension 128, input length 8, and token length 1. The weighting hyperparameters (α, β) for the auxiliary losses $(\mathcal{L}_{ca}, \mathcal{L}_{rec})$ are tuned per dataset: (0.5, 0.1) for MOSI, (0.9, 10) for MOSEI, and (0.9, 0.1) for SIMS.

4.4. Comparison with State-of-the-art Methods

We compare DNRNet against several strong baselines, including MISA [8], Self-MM [38], MMIM [7], CENET [28], TETFN [27], TFR-Net, ALMT [44], LNLN [45], and P-RMF [48]. The results of these baselines are reported as in LNLN, ensuring consistency in evaluation settings across all compared methods. As shown in Tables 1 and 2 (averaged across all missing rates), DNRNet consistently achieves state-of-the-art performance. Specifically, DNRNet consistently outperforms all existing methods across all metrics on the MOSI dataset. Notably, it achieves the lowest MAE of 1.036, while its Acc-2 and F1 scores show improvements of 1.66% and 1.59%, respectively, over the next best model, P-RMF. Furthermore, DNRNet demonstrates a strong advantage on the larger MOSEI dataset, particularly achieving the highest accuracy for both Acc-7 (46.89%) and Acc-5 (47.84%) metrics. On the Chinese multimodal dataset SIMS, DNRNet surpasses the second-best model on almost all metrics, providing strong evidence of its robust performance under frame-level missingness.

Furthermore, Figure 3 depicts the performance trends of

Table 2. Comparison of model performance on SIMS. Note: The lower MAE corresponds to superior results.

Method	Acc-5	Acc-3	Acc-2	F1	MAE	Corr
MISA	31.53	56.87	72.71	66.30	0.539	0.348
Self-MM	32.28	56.75	72.81	68.43	0.508	0.376
MMIM	31.81	52.76	69.86	66.21	0.544	0.339
CENET	22.29	53.17	68.13	57.90	0.589	0.107
TETFN	33.42	56.91	73.58	68.67	0.505	0.387
TFR-Net	26.52	52.89	68.13	58.70	0.661	0.169
ALMT	20.00	45.36	69.66	72.76	0.561	0.364
LNLN	34.64	57.14	72.73	79.43	0.514	0.397
P-RMF	34.83	54.75	73.64	74.65	0.500	0.414
DNRNet	35.71	59.54	73.82	77.50	0.498	0.416

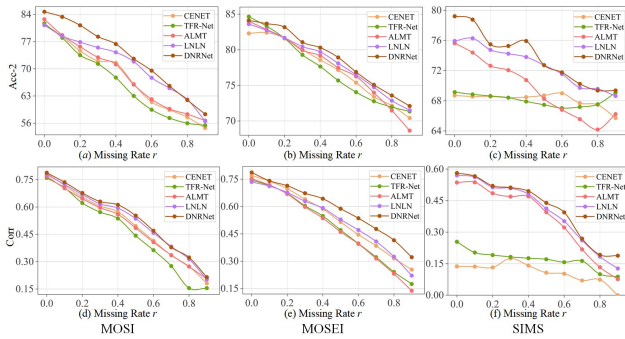


Figure 3. Visualization of performance under different missing rate (Acc-2 $\uparrow$ and Corr $\uparrow$).

Accuracy and Corr for DNRNet and several representative baselines across increasing missing rates. We observe that, as the missing rate increases, the performance of all baseline models exhibits a steep decay. This indicates their difficulty in effectively generating high-quality feature representations when faced with severely incomplete input features. In sharp contrast, our proposed model maintains significant superiority and stability in its performance metrics, even under high missing rate conditions. This further validates the efficacy and advantages of the dynamic nested recurrent mechanism. Detailed performance in different missing rates can be found in the Supplementary.

4.5. Ablation Studies

Contributions of Different Components. To assess the contribution of each component within DNRNet, we conducted a series of ablation studies by removing key modules and observing the resulting performance changes, as shown in Table 3. The results indicate that performance significantly degrades when the recurrent process is limited to a single iteration (w/o multi-round recurrence), highlighting that the multi-step iterative pattern completion process is critical for accurately inferring and correcting incomplete

features. This goal cannot be fully achieved through a single feed-forward computation. Removing the local perception module (w/o Local Perception) breaks the inner loop and prevents the model from generating local correction features, leading to performance degradation. This is because the model is unable to actively infer and restore local semantic information based on the prior expectations of the modality itself, thereby reducing its self-completion capability. Furthermore, removing the global perception module (w/o Global Perception) breaks the outer recurrence loop, weakening the model’s ability to dynamically extract global correction features from complementary modalities for cross-modal completion. This indicates that the nested design and active perception-inference mechanisms collectively form a robust framework for handling complex missing data. In conclusion, every component plays an indispensable role in processing complex frame-level missing data for sentiment analysis, validating the effectiveness of the proposed dynamic nested recurrent network.

Contributions of Different Losses. We evaluated the contribution of each loss component to the performance of DNRNet, as presented in Table 3. The experimental results indicate that the exclusion of these losses leads to a noticeable degradation in performance. Removing the confidence-aware loss ($\mathcal{L}_{ca}$) significantly weakens the model’s ability to dynamically perceive modality disparity, which is crucial for robust multimodal fusion and accurate data completion in the task of intra-modality frame-level data missingness. Furthermore, the absence of the Reconstruction loss ($\mathcal{L}_{rec}$) compromises the model’s ability to reconstruct the missing semantic content, consequently increasing the semantic deviation between the completed features and the original complete features. These results suggest that each loss component plays a complementary and indispensable role in the optimization process, collectively guiding DNRNet to learn more precise and missingness-robust feature representations for the frame-level data missingness sentiment analysis task.

4.6. Classification Performance Visualization

To evaluate the classification performance of the proposed method and its robustness to data missingness, we visualized the confusion matrices of the baseline model LNLN and our DNRNet on the MOSI dataset for missing rates of 0.0, 0.1, 0.5, and 0.9. As shown in Figure 4, both models exhibit category prediction bias as the missing rate increases, reflecting the significant challenge of extracting highly discriminative emotional features under frame-level data missingness. For instance, when the missing rate is 0.9, the prediction of the baseline model LNLN is almost entirely concentrated on the neutral category. This indicates that when facing severe missingness of modal information, the model will trivially classify all samples as belonging to the dom-

Table 3. Effects of different components and losses. Lower MAE indicates superior results.

Method	MOSI						SIMS					
	Acc-7	Acc-5	Acc-2	F1	MAE	Corr	Acc-5	Acc-3	Acc-2	F1	MAE	Corr
w/o multi-round recurrence	34.94	38.86	73.95 / 72.57	73.68 / 72.34	1.058	0.520	35.21	58.89	73.54	77.01	0.505	0.393
w/o Local Perception	34.76	38.59	73.66 / 72.41	73.72 / 72.45	1.049	0.522	34.67	58.45	73.18	76.83	0.518	0.378
w/o Global Perception	34.57	38.36	73.29 / 72.22	73.46 / 72.25	1.059	0.518	34.61	58.40	73.05	76.60	0.521	0.350
w/o $\mathcal{L}_{ca}$	34.05	37.62	72.88 / 71.44	72.91 / 71.52	1.072	0.510	34.08	58.57	73.21	76.94	0.516	0.377
w/o $\mathcal{L}_{rec}$	34.34	38.39	73.43 / 72.11	73.28 / 71.86	1.051	0.529	34.11	58.58	73.40	77.00	0.508	0.382
DNRNet	35.36	39.11	74.22 / 73.19	74.24 / 73.28	1.036	0.538	35.71	59.54	73.82	77.50	0.498	0.416

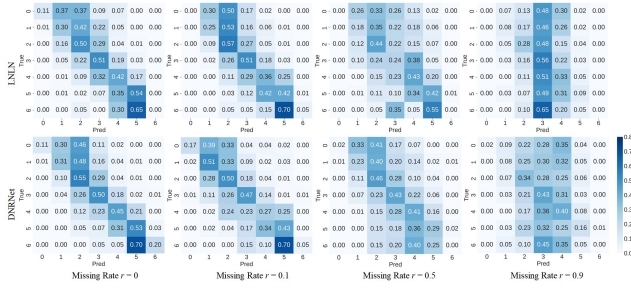


Figure 4. Confusion matrices of DNRNet and the baseline model LNLN on the MOSI dataset. Note: 0-6 denote strongly negative, negative, weakly negative, neutral, weakly positive, positive, and strongly positive, respectively.

inant category. Consequently, traditional feedforward networks are unable to recover sufficient semantic information from highly incomplete inputs, leading to the collapse of semantic distinguishability and subsequent classification failure. In contrast, the prediction distribution of our proposed DNRNet remains dispersed across multiple emotional categories even at a high missing rate, demonstrating its superior discriminative capability. We attribute this advantage to the fact that DNRNet is capable of continuously inferring and correcting the damaged modal representations through both local and global features. This process ensures that the ultimately fused features are more distinguishable in the semantic space, effectively preventing the degradation of classification boundaries.

4.7. Feature Visualization

To intuitively illustrate the effectiveness of our proposed modules, we provide t-SNE feature visualizations and reconstruction error boxplots under a missing rate of 0.7. As illustrated in Figure 5(a), compared to Incomplete Input, the ablation variants (w/o Global and w/o Local) represent data that is closer to Complete data, but they still exhibit significant distribution differences. In contrast, the reconstructed features of the Full DNRNet are tightly clustered and highly overlap with the Complete data, exhibiting a similar semantic manifold. This proves its superior feature recovery capa-

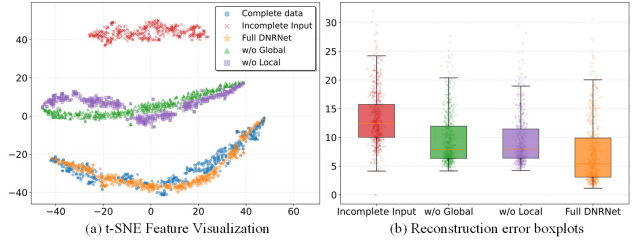


Figure 5. The t-SNE feature visualization and reconstruction error boxplots under a missing rate of 0.7.

bility. To provide quantitative support, Figure 5(b) presents the reconstruction error boxplots. Compared to the Full DNRNet, removing either the local or global loop leads to a significant increase in error, demonstrating that both loops are indispensable for generating high-quality features.

5. Conclusion

In this paper, we propose the Dynamic Nested Recurrent Network (DNRNet) to address the frame-level missingness in multimodal sentiment analysis. The core nested architecture is inspired by the brain’s perceptual inference, modeling both the intra-cortical pattern completion mechanism and the corticothalamic circuit. Specifically, the local recurrent loop recovers intra-modality semantics by generating local correction features based on prior expectations, while the global recurrent loop dynamically integrates cross-modal information and generates global correction features. Both loops collaboratively engage in iterative refinement, endowing DNRNet with a powerful self-adaptive correction capability for eliminating redundancy and noise. Extensive experiments confirm that DNRNet significantly outperforms state-of-the-art baselines, validating the effectiveness of our approach. For future work, we plan to address the dual challenge of missing data and class imbalance, which frequently co-occur in imbalanced datasets. We will investigate integrating imbalanced learning techniques within the recurrent mechanism to mitigate model bias toward majority classes.

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